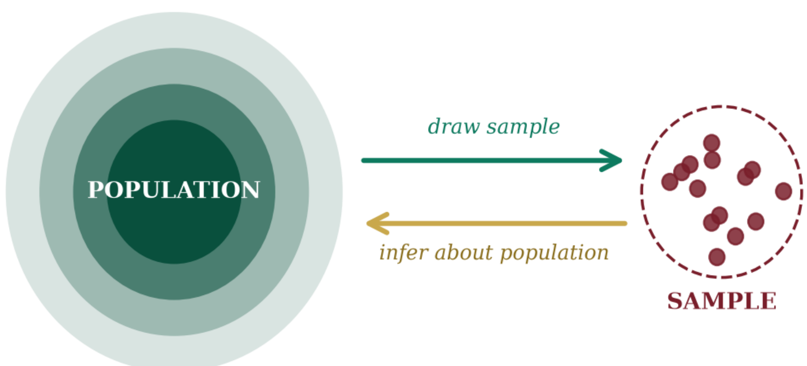
— STATISTICAL FOUNDATIONS —

Task Checklist

Part A · Theoretical Foundation

A concise conceptual companion to inferential statistics — covering sampling, hypothesis testing, confidence, error, and the relationships between variables.



INFERENCE · PROBABILITY · EVIDENCE

Q1. What is Inferential Statistics?

DEFINITION

Inferential statistics is the branch of statistics that uses information drawn from a sample to make reasoned conclusions, predictions, and generalisations about a wider population. Where descriptive statistics merely summarises observed data, inferential statistics extends the reach of that data — accepting a measurable degree of uncertainty in exchange for insight beyond what was directly observed.

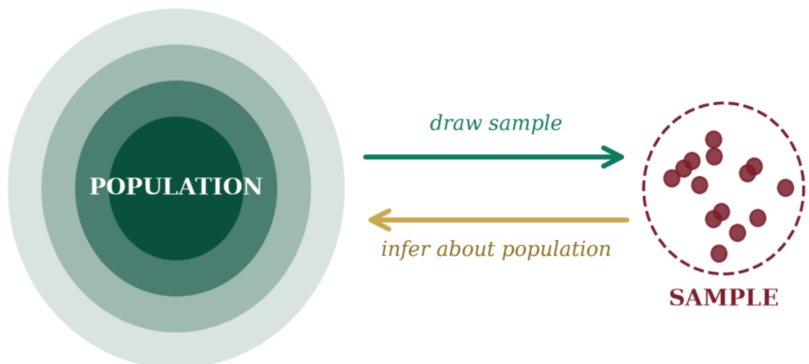


FIG. 1 — *A sample is drawn from a population; conclusions then flow back.*

Q2. Hypothesis Testing & Its Components

CONCEPT

Hypothesis testing is a structured statistical procedure for deciding whether sample evidence is strong enough to support a claim about a population. It converts an investigative question into two competing statements and then uses probability to choose between them.

CORE COMPONENTS

- Null hypothesis (H_0) — the default assumption of no effect or no difference.
- Alternative hypothesis (H_1) — the claim the researcher seeks to support.
- Significance level (α) — the acceptable risk of a false rejection, often 0.05.
- Test statistic — a value computed from the sample (z, t, χ^2, F).
- p-value & decision rule — the basis for accepting or rejecting H_0.

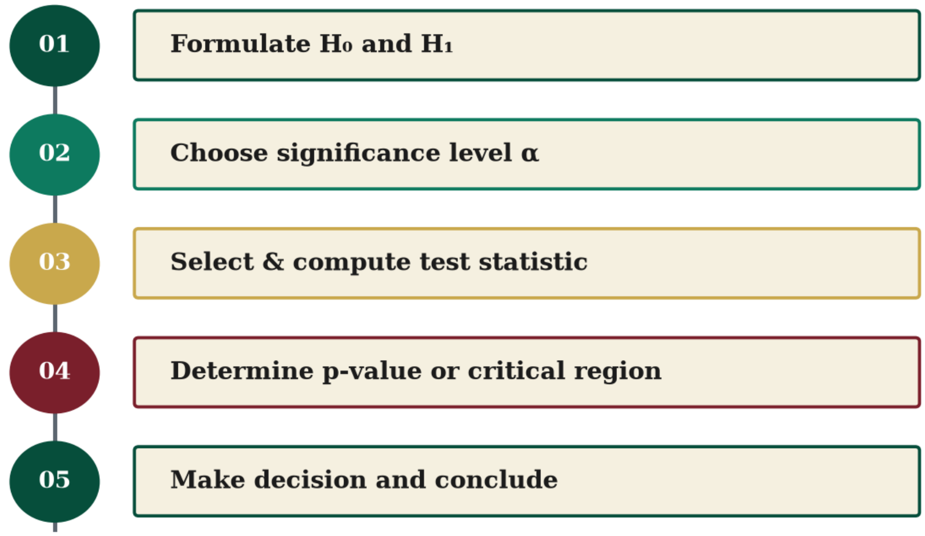


FIG. 2 — *The five-stage hypothesis-testing pipeline.*

Q3. Confidence Interval & Critical Value

CONFIDENCE INTERVAL

A **confidence interval** is a range of values, computed from the sample, that is likely to contain the true population parameter with a stated probability — typically 95%. It expresses both an estimate and the uncertainty surrounding it.

CRITICAL VALUE

The **critical value** is the threshold on the sampling distribution that separates the acceptance region from the rejection region. For a two-tailed test at 95% confidence under a normal distribution, the critical values are ±1.96.

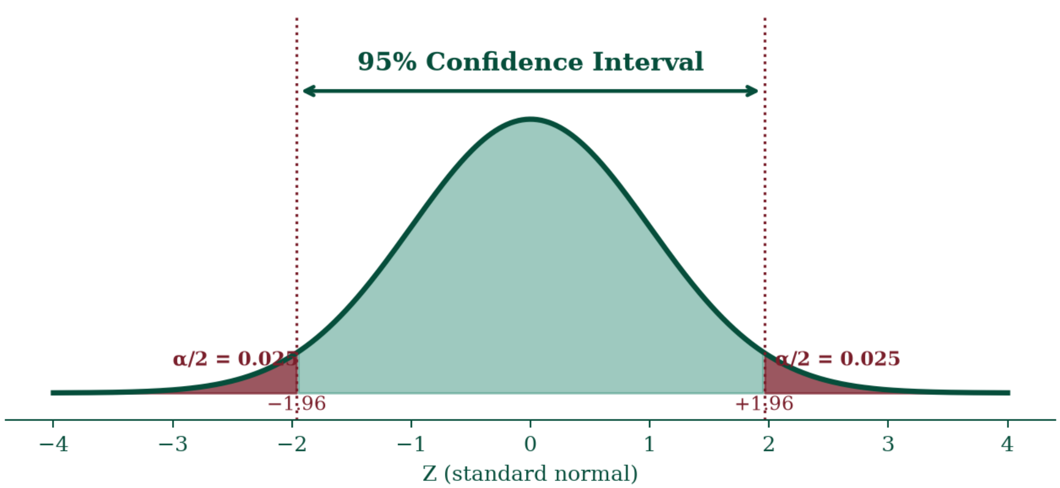


FIG. 3 — *95% confidence interval bounded by the critical values ±1.96.*

Q4. The p-value

DEFINITION

The p-value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. A small p-value indicates that the observed data would be unlikely under H_0, providing evidence to reject it; a large p-value suggests the data are consistent with H_0.

> **DECISION RULE**
>
> *If $p \le \alpha$, reject H_0 (the result is statistically significant). If $p > \alpha$, fail to reject H_0 (insufficient evidence to support H_1).*

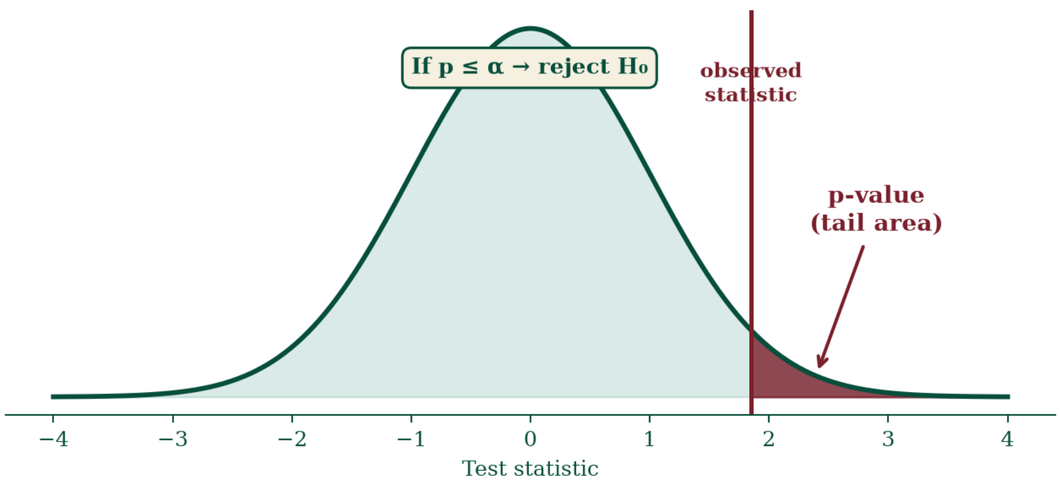


FIG. 4 — *The p-value is the tail probability beyond the observed test statistic.*

Q5. Type I and Type II Errors

TWO WAYS TO BE WRONG

Type I error (α) — rejecting a null hypothesis that is actually true. Conceptually, a “false alarm.”

Type II error (β) — failing to reject a null hypothesis that is actually false. Conceptually, a “missed detection.”

For a fixed sample size, reducing one type of error generally inflates the other; increasing sample size is the principal way to reduce both.

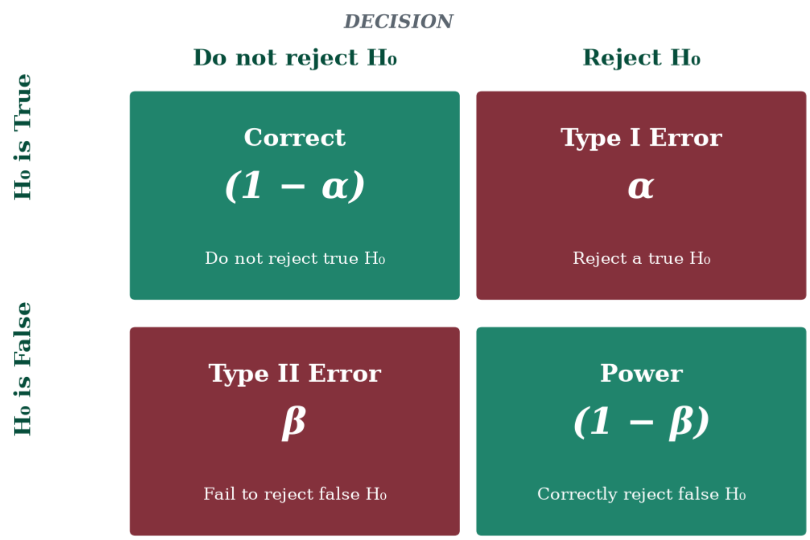


FIG. 5 — *The decision matrix: correct outcomes and the two error types.*

Q6. Z-test, T-test, Chi-square and ANOVA

FOUR FUNDAMENTAL TESTS

Z-test — compares a sample mean to a population mean when the population standard deviation is known and the sample is large ($n \geq 30$).

T-test — used when the population standard deviation is unknown and samples are small; variants include one-sample, two-sample independent, and paired.

Chi-square (χ^2) test — applied to categorical data to test goodness-of-fit or independence between two variables in a contingency table.

ANOVA — (Analysis of Variance) tests whether the means of three or more groups differ significantly by comparing between-group and within-group variance.

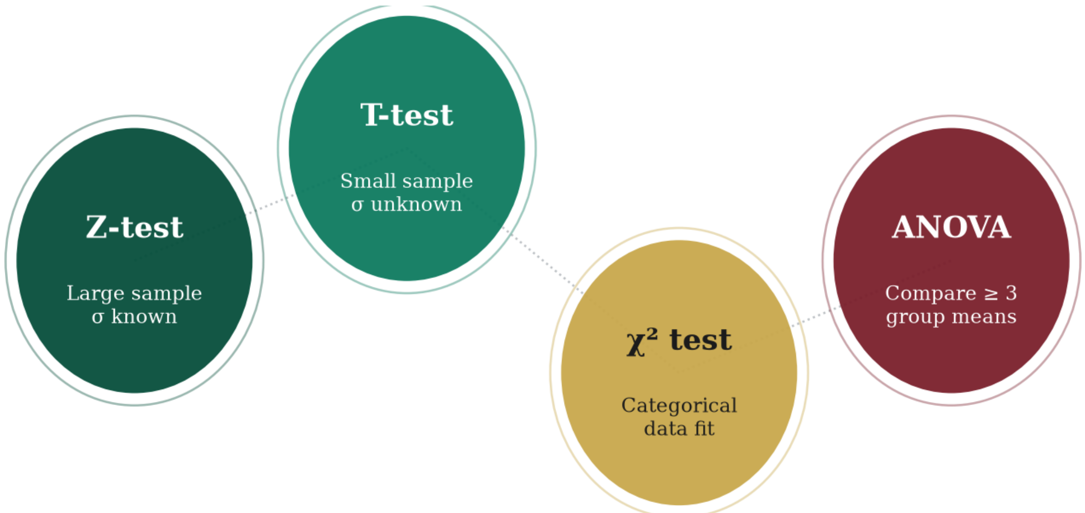


FIG. 6 — *Choosing the right test depends on sample size, σ, and data type.*

Q7. Covariance

DEFINITION

Covariance measures the direction of the linear relationship between two variables. A positive value indicates that the two variables tend to move together; a negative value indicates they move in opposite directions; a value near zero suggests no linear relationship. Because its magnitude depends on the units of the variables, covariance is not directly comparable across datasets.

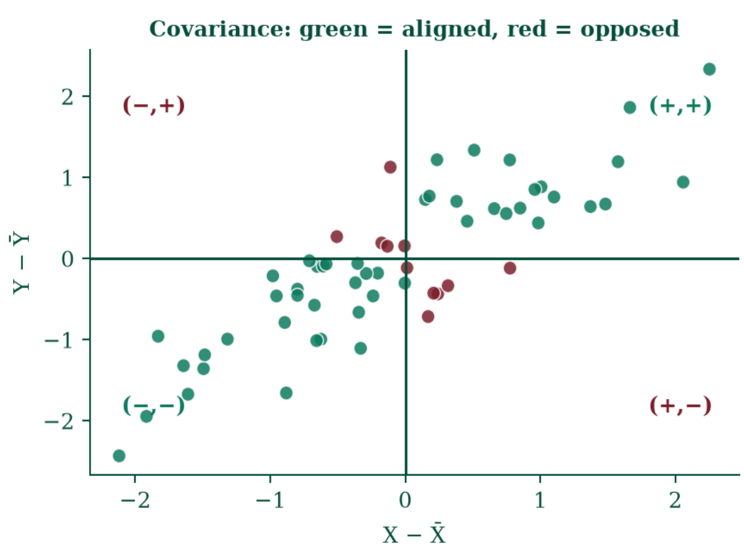


FIG. 7 — *Quadrant view: green points strengthen, red points weaken covariance.*

Q8. Correlation

DEFINITION

Correlation is a standardised measure of the strength and direction of a linear relationship between two variables. The Pearson correlation coefficient (r) always lies between −1 and +1, where +1 denotes a perfect positive relationship, −1 a perfect negative one, and 0 no linear relationship. Unlike covariance, correlation is unitless and directly comparable across different pairs of variables.

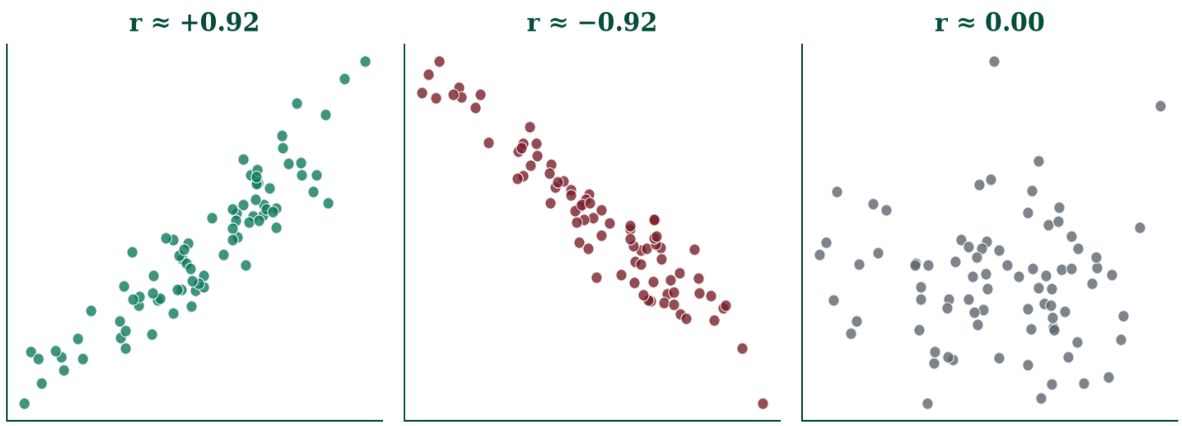


FIG. *8 — Strong positive, strong negative, and no linear correlation.*

IN SUMMARY

Inferential statistics turns limited sample evidence into population-level conclusions. Hypothesis testing, confidence intervals, p-values, and the disciplined choice of test (z, t, χ^2, ANOVA) form its working toolkit, while covariance and correlation describe how variables move in concert.